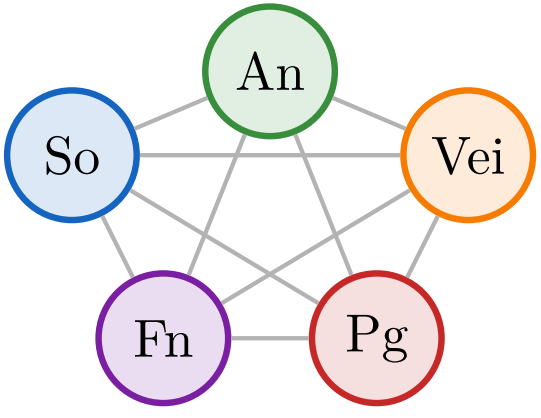
Living material



variational multiphase model
15 unknown interactions a_{ij}

Bayesian calibration

composition

+ viability

TMCMC on a GPU (JAX)
3 seeds per stage, explicit convergence and identifiability checks

What the data determine

5–9 of 15 interactions constrained

in silico:
without *F. nucleatum*, no late rise of *P. gingivalis*